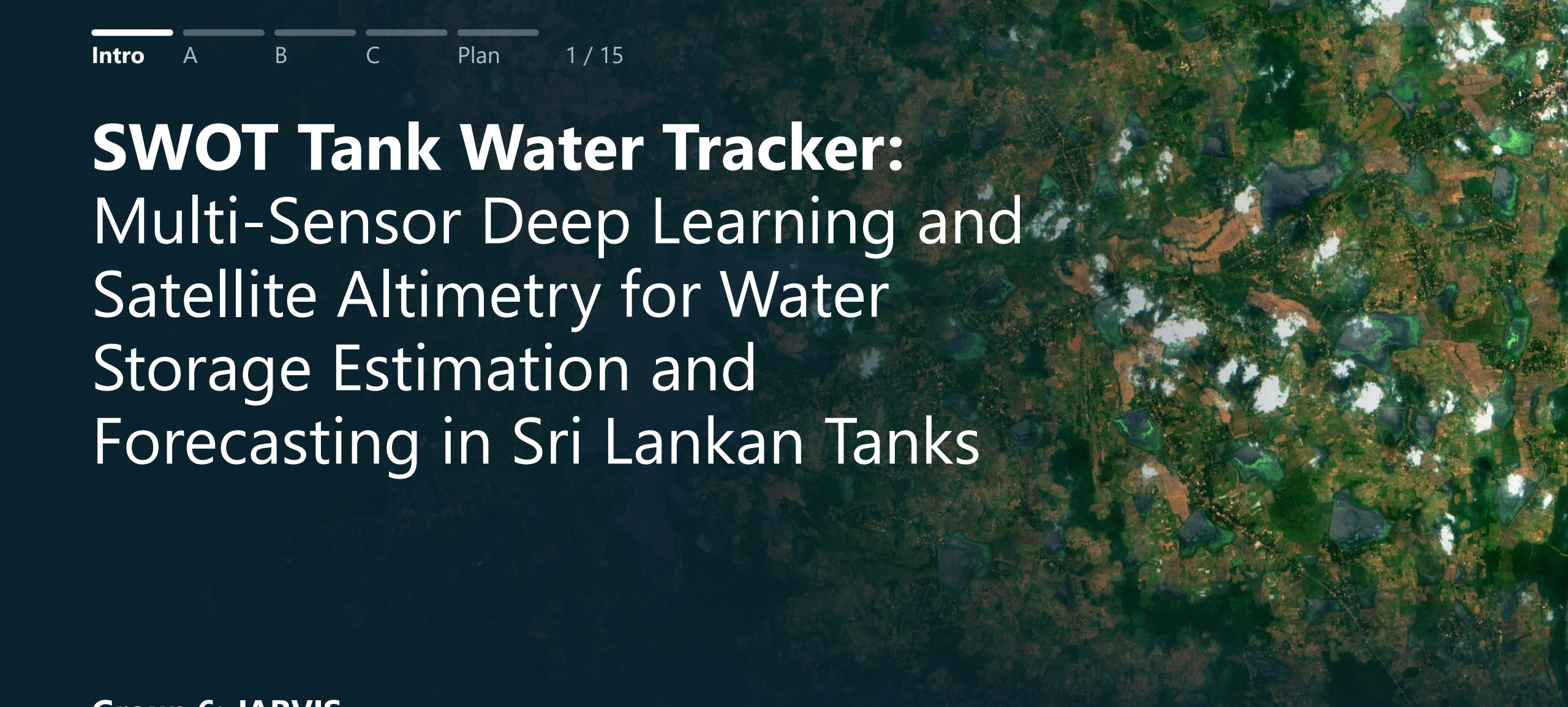
SWOT Tank Water Tracker: Multi-Sensor Deep Learning and Satellite Altimetry for Water Storage Estimation and Forecasting in Sri Lankan Tanks



Group 6: JARVIS

SUPERVISORS Dr.(Ms) Ganegoda G.U. (IT) Prof.(Mrs) Sandanayake T.C. (IDS)

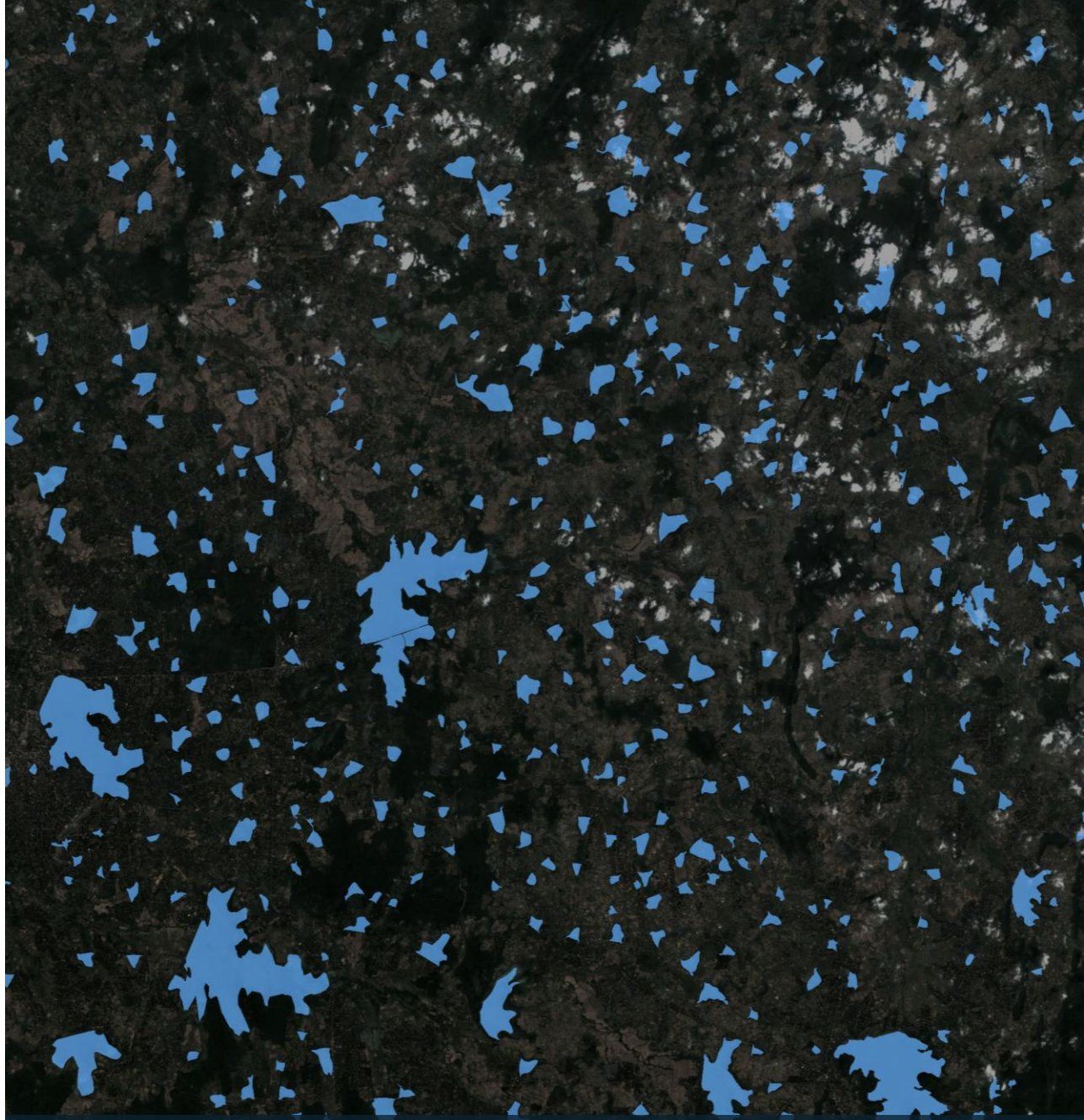
Tanks around Mahakanadarawa, Sentinel-2, 28 February 2026. Contains modified Copernicus Sentinel data 2026.

Problem and Research Gap

- Only **large reservoirs** have storage figures (Daily FT 2023; Irrigation Dept. 2026)
- **Village tanks** have no gauges or records
- Satellite maps give **area, not volume** (IFAD 2026)

RESEARCH GAP

- ICESat-2: heights every **91 days** (Madumalsha et al. 2025)
- SWOT storage: tested **abroad only** (Das & Hossain 2025; Ma et al. 2026)



432 tanks of 6.25 ha or more on this image (blue). Sentinel-2, 28 February 2026. Contains modified Copernicus Sentinel data 2026; tank outlines © OpenStreetMap contributors.



Illustration: NASA/JPL-Caltech.

Aim and Research Questions

Aim: estimate and forecast storage, including ungauged village tanks.

RQ1
Water maps under cloud and plants

RQ2
SWOT accuracy and “suspect” readings

RQ3
% full for unseen tanks

RQ4
2 to 4-week storage forecast

Study Area and Data

5

major reservoirs, daily gauges

8

village tanks near Mahakanadarawa

FREE DATA

- SWOT heights
- Sentinel-1 and -2 images
- Daily storage, **117** reservoirs

  (Irrigation Dept. 2026)
- Water areas, **about 70** reservoirs

  (Global Water Watch 2026)
- Rain and evaporation



Our own figure (Group 6, JARVIS). The study area map uses © OpenStreetMap contributors.

How It Works: One Chain, Three Modules

A



AI water map

Owner: Member A

B



SWOT heights and storage

Owner: Member B

C



Forecast and system

Owner: Member C

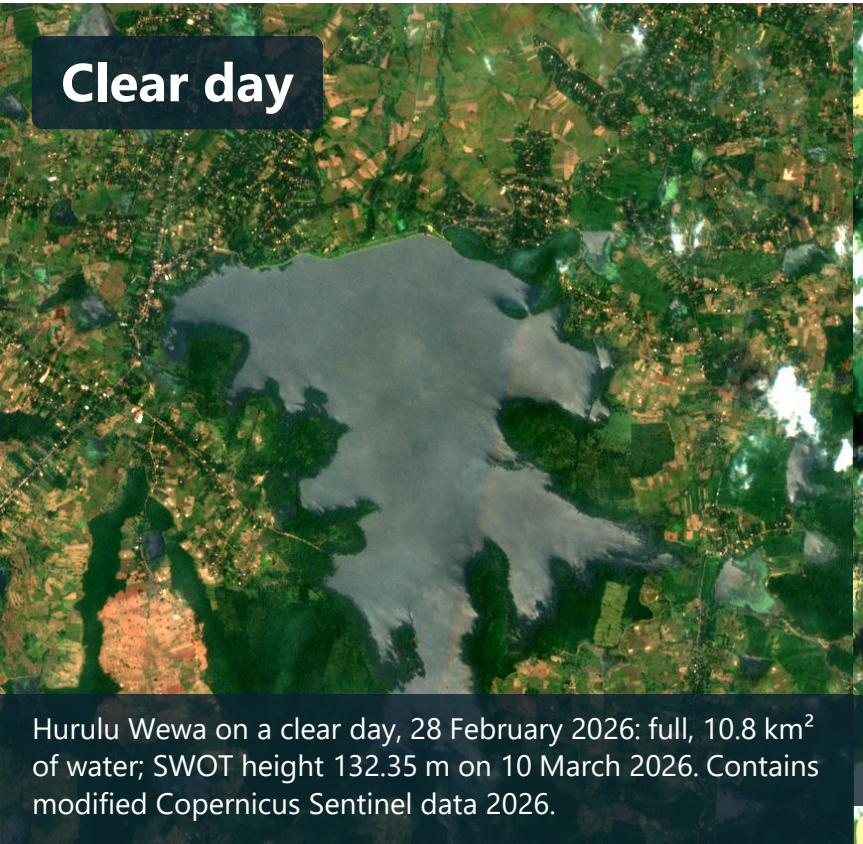
Storage every **5 to 12 days**.

Related Work and Gap

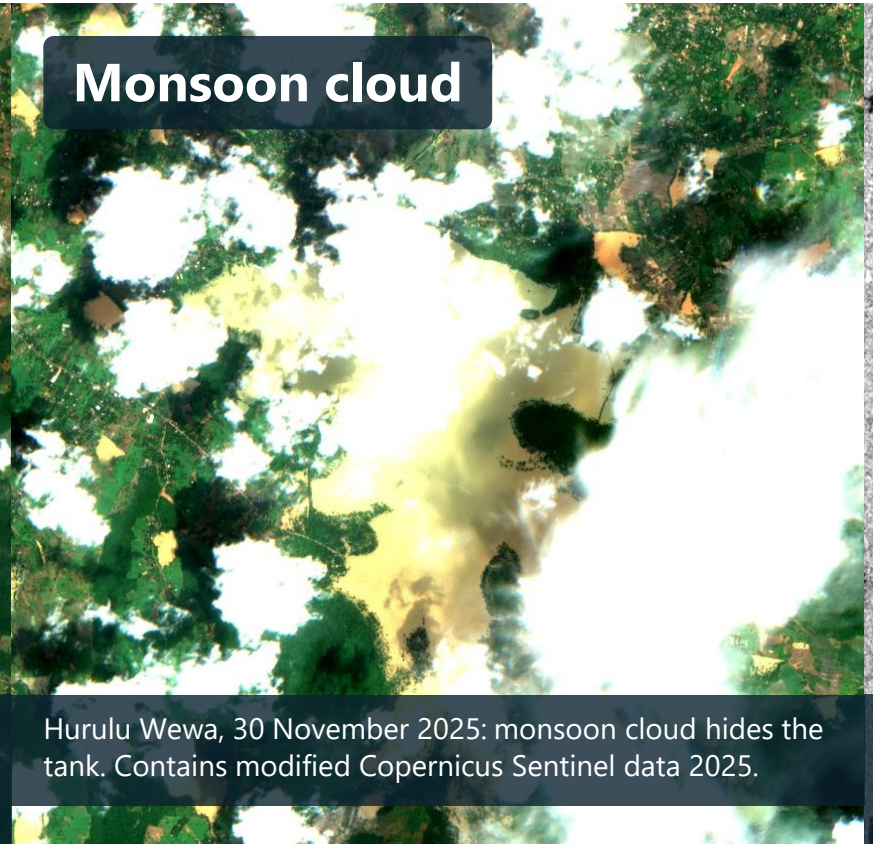
Owner: Member A

- Sentinel-1: **R^2 0.67 to 0.94** (Perera & Jayaweera 2020)
- U-Net, **no SWOT** (Pruekthikanee 2026)

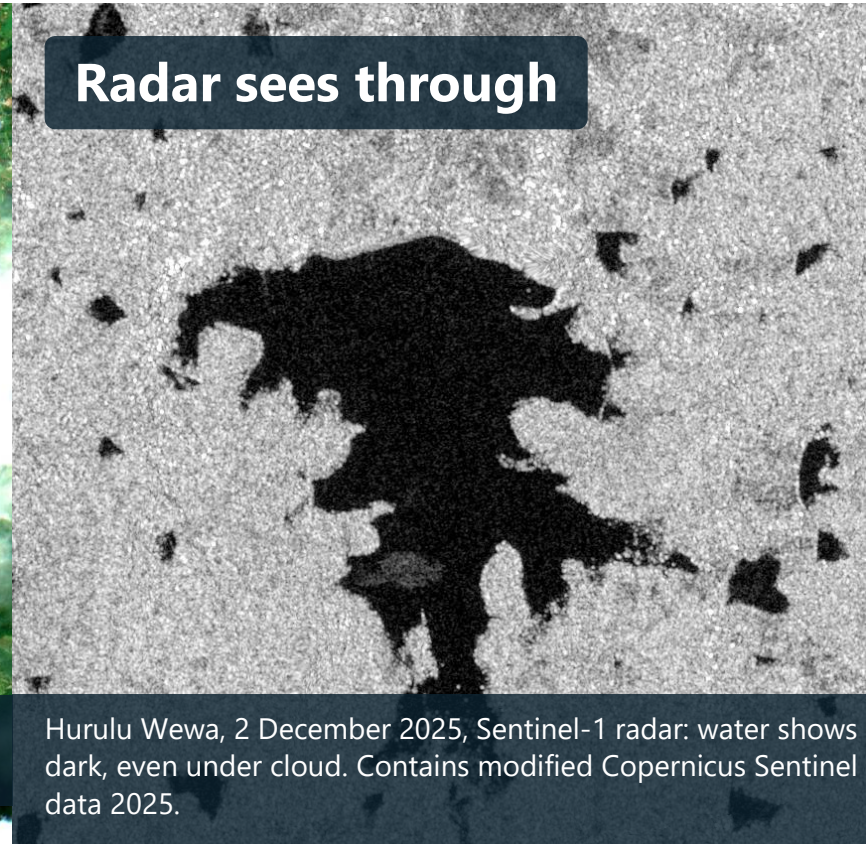
Clouds and floating plants hide the water (Dawalagala et al. 2023). In our check, only **14 to 22%** of Sentinel-2 dates were clear.



Hurulu Wewa on a clear day, 28 February 2026: full, 10.8 km² of water; SWOT height 132.35 m on 10 March 2026. Contains modified Copernicus Sentinel data 2026.



Hurulu Wewa, 30 November 2025: monsoon cloud hides the tank. Contains modified Copernicus Sentinel data 2025.



Hurulu Wewa, 2 December 2025, Sentinel-1 radar: water shows dark, even under cloud. Contains modified Copernicus Sentinel data 2025.

Approach and Data

Owner: Member A

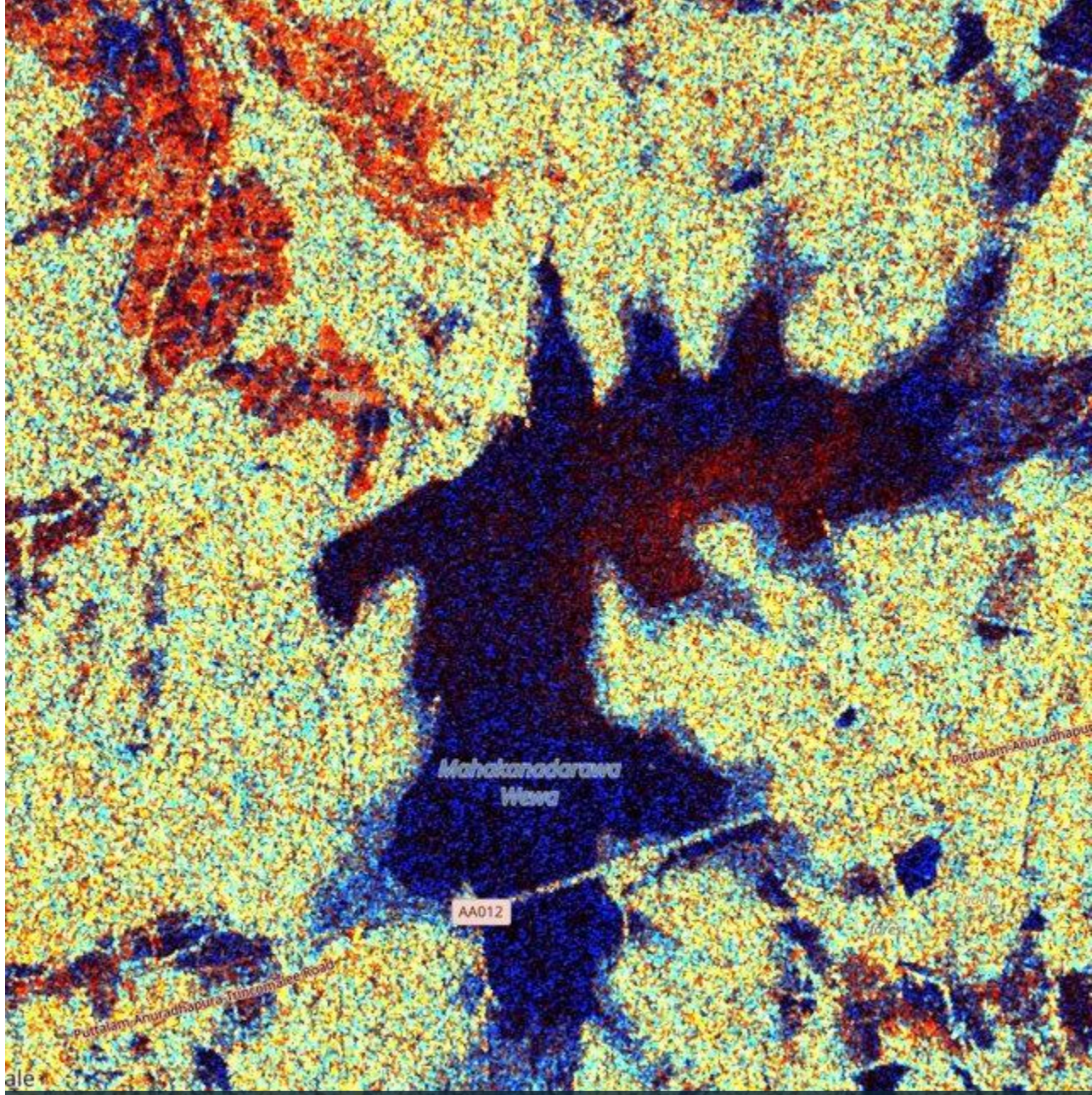
U-Net on **Sentinel-1 radar** + **Sentinel-2 photos**: maps water under cloud

THREE CLASSES

Open water **Water under plants** **Land**

Pre-trained on Sen1Floods11 (Bonafilia et al. 2020) and S1S2-Water (Wieland et al. 2024)

Fine-tuned on our masks, including **cloudy** and **plant-covered** tanks



Mahakanadarawa on 26 November 2025, as Cyclone Ditwah formed, when photos showed only cloud. Open water is dark blue. Contains modified Copernicus Sentinel data 2025; labels © OpenStreetMap contributors.

Evaluation and Early Evidence

Owner: Member A

- **IoU** and **F1** against hand labels
- Cloudy and plant-covered tanks scored separately
- Baselines: simple water-index and radar-threshold rules

57-100%

how far off simple rules were at **Kanhindawewa**, a plant-covered tank



Why we add our own labels: Bulankulama tank, 25 March 2026, covered in plants. A simple water index misses it, yet SWOT has 47 usable heights here. Contains modified Copernicus Sentinel data 2026.

SWOT Heights Against the Gauges

Owner: Member B

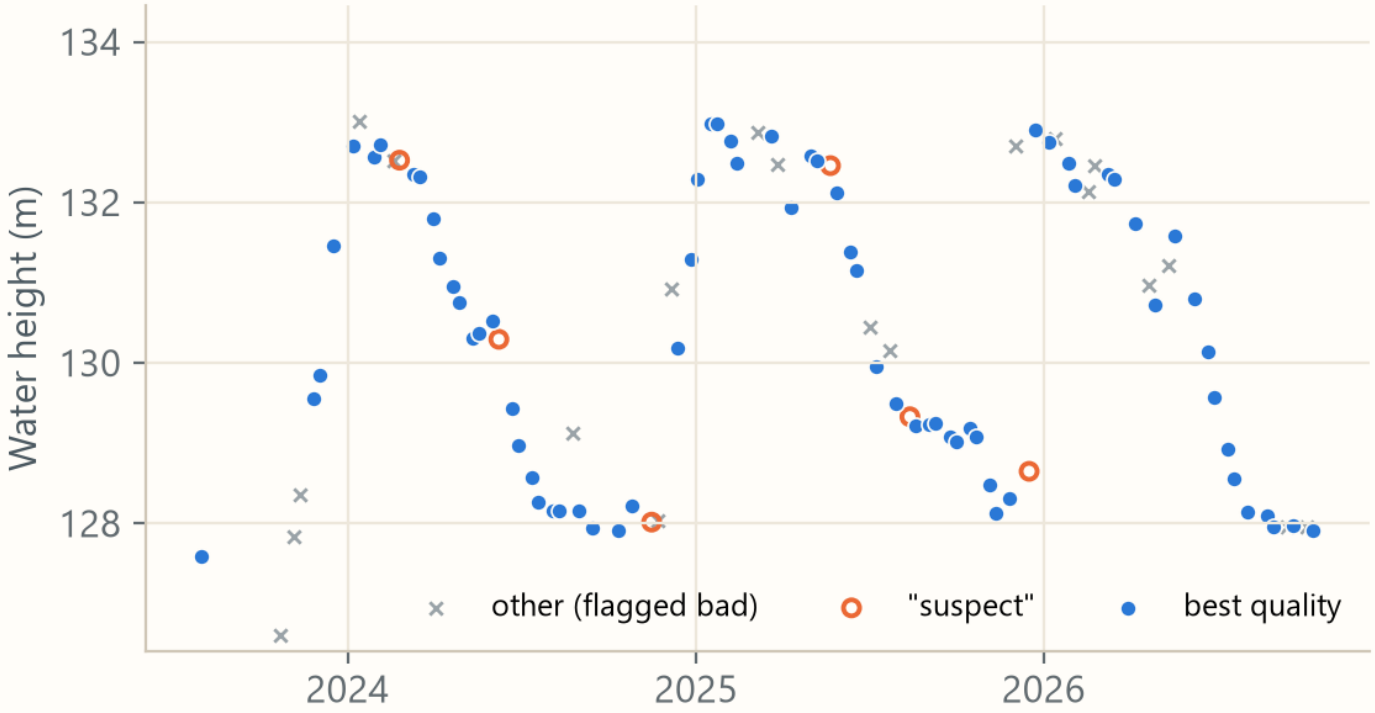
AT THE 5 MAJOR RESERVOIRS

0.40–0.64 m

fixed offset above the department's levels, removed once per reservoir

0.07–0.13 m

scatter around it



Hurulu Wewa: every SWOT pass, July 2023 to October 2026 (101 readings; 4 far-off ones are off the chart).
Data: NASA/CNES SWOT, via Hydrocron.

47 to 90 usable readings per tank

Trust Screen for “Suspect” Readings

Owner: Member B

SWOT marks almost every small-tank reading **“suspect”**.

Screen: **gradient boosting** on SWOT’s quality fields.

AVERAGE ERROR



Storage Model: Height + Area → % Full

Owner: Member B

LightGBM, about 100 gauged reservoirs

ERROR, UNSEEN RESERVOIRS (% OF CAPACITY)

8%	11%	11%
model	under 2 km²	formula



Hurulu Wewa, 29 July 2023: low, 6.1 km² of water; SWOT height 127.59 m on 30 July 2023. Contains modified Copernicus Sentinel data 2023.

Hurulu Wewa, 30 July 2025: half full, 8.3 km² of water; SWOT height 129.50 m the same day. Contains modified Copernicus Sentinel data 2025.

Hurulu Wewa on a clear day, 28 February 2026: full, 10.8 km² of water; SWOT height 132.35 m on 10 March 2026. Contains modified Copernicus Sentinel data 2026.

Related Work and Approach

Owner: Member C

UoM forecasts used gauges (Meyen 2018); village tanks have none.

INPUTS

- Storage
- Season
- Rain
- Evaporation

MODELS

- **LSTM**
- Gradient boosting
- “No change”

Storage at 1, 2, 4 weeks

Trained on **117 reservoirs** (Irrigation Dept. 2026)

Evaluation and Early Evidence

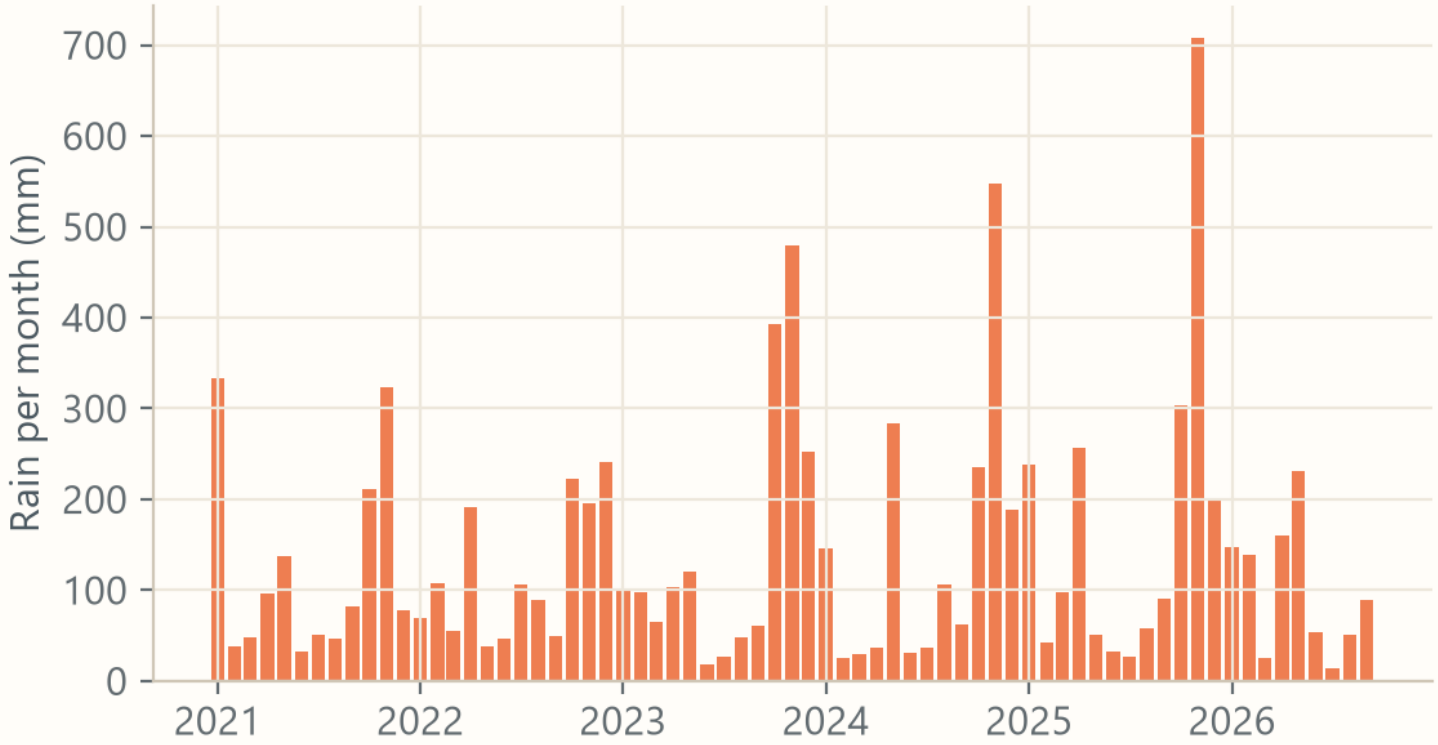
Owner: Member C

13-22%

better than “no change” (gradient boosting)

NEXT

- Test the LSTM on unseen reservoirs
- Error in % of capacity at 2 and 4 weeks

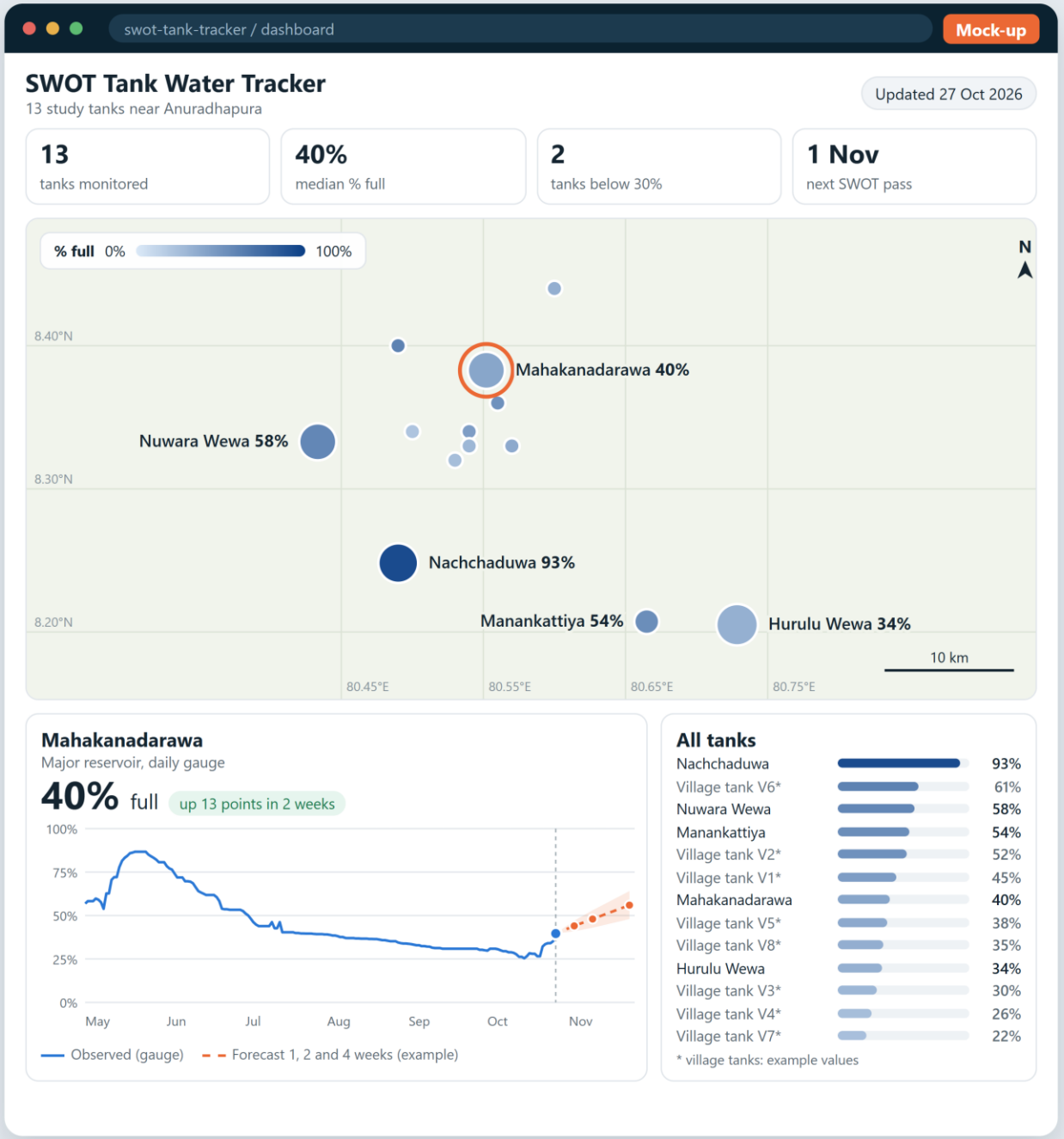


Monthly rain at Hurulu Wewa, January 2021 to September 2026: the October to January rains fill the tanks. Data: ERA5 via Open-Meteo (CC BY 4.0).

System and Dashboard

Owner: Member C

- Logger running since **26 Oct 2026**
- Runs A → B → C after each pass
- **FastAPI** back end
- **Next.js + Leaflet** dashboard
- For irrigation officers and farmer organisations



Mock-up of the planned dashboard. Major reservoirs: Irrigation Department daily storage, 27 Oct 2026. Village tanks and the forecast: example values.

Plan of Action

- **Oct-Nov 2026** Proposal and talk
- **15 Dec 2026** Study area confirmed
- **Jan-Apr 2027** The three models
- **Mar-May 2027** Checks and dashboard
- **30 Jun 2027** Thesis and paper

Thank you. Questions?

The same view, darkened. Sentinel-2, 28 February 2026. Contains modified Copernicus Sentinel data 2026.